# Kepler’s laws

**antocciphysics**



**Name:** _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

**LEARNING TARGET** Use orbital shape, equal areas and period comparisons.

**SUCCESS CRITERIA - I can...**

1. Place the central body at a focus of an ellipse.
2. Compare speed near and far from the central body.
3. Use $T^2=a^3$ only with the specified units and central body.

**Core:** teacher selects from 1-7. **Stretch:** 8 after checking your work.

**MARZANO SELF-CHECK**

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

## 01 Warm-up: bring back an earlier idea

A launch has $v_x$=5 m/s and $v_y$, i=30 m/s upward. Use g=10 m/s². Find time to the top and total time back to launch height.

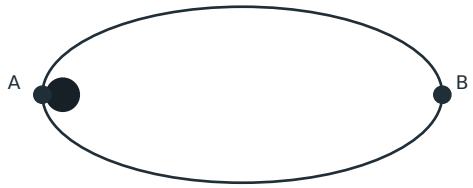
**Marzano check - circle:** ① ② ③

## 02 Worked example: explain the model

**Reference:** $T^2 = a^3$.
For Sun orbits only when T is in years and a is in AU. a is the semi-major axis. Equal areas are swept in equal times.

**Worked reasoning:** For an ideal Sun orbit, a=4 AU gives $T=\sqrt{(4^3)}=8$ years. The orbit period depends on semimajor axis, not on a chosen point's instantaneous speed alone.

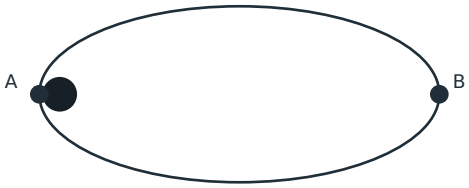
**Your turn:** Draw an ellipse, put the Sun at a focus and mark the faster and slower parts.




**Marzano check - circle:** ① ② ③

## 03 Read and interpret the model

Mark the Sun at a focus on an ellipse. Label where the object is fastest and slowest.




**Marzano check - circle:** ① ② ③

## 04 Apply the physics

Supplied comparisons: a=1 AU gives T=1 year; a=4 AU gives T=8 years. Which orbit takes longer? How many Earth years for two orbits of the outer object?

| Semimajor axis (AU) | Period (years) |
|---|---|
| 1 | 1 |
| 4 | 8 |

**Marzano check - circle:** ① ② ③

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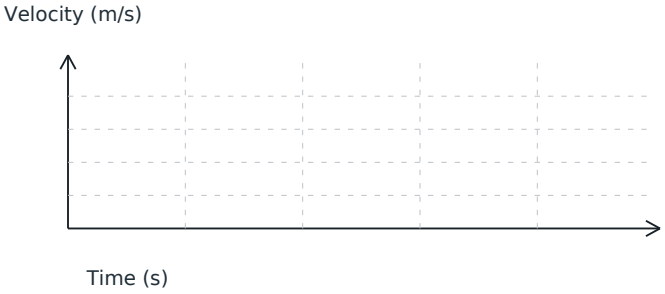
## 05 Explain your reasoning

Does knowing an orbit’s period alone tell the date when an asteroid reaches Earth’s location? What else is needed?

**Marzano check - circle:** ① ② ③

## 06 Review: connect an earlier idea

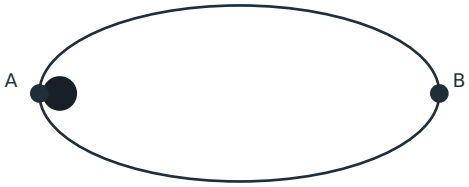
A ball is dropped from rest and falls ideally for 5 s. Use $g=10\ m/s^2$. Find its speed and direction.
Sketch the motion and label a positive direction.




**Marzano check - circle:** ① ② ③

## 07 Exit check

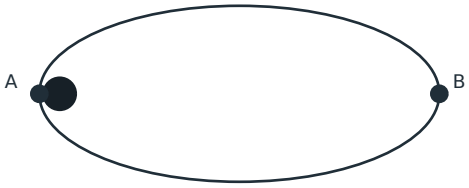
An asteroid is nearest the Sun at point A and farthest at B. At which point is it fastest? Name which law helps explain this.




**Marzano check - circle:** ① ② ③

## 08 STRETCH: test the model

Two Sun orbits have supplied periods 1 and 8 years. Their semimajor axes are 1 and 4 AU. Explain why four times the size does not mean four times the period.




**Marzano check - circle:** ① ② ③